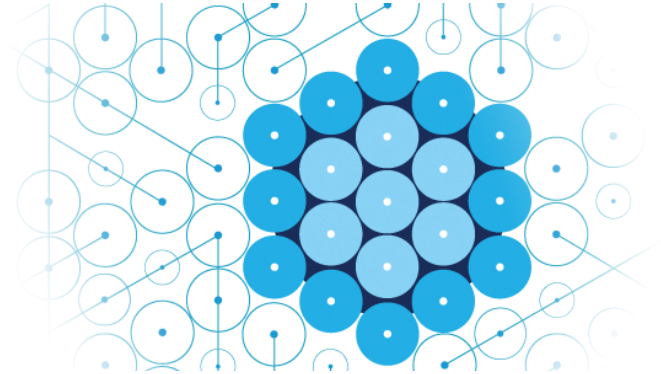


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# **HR Analytics Dashboard Documentation** **IBM HR Employee Attrition Analysis using Power BI**

# **1. Project Overview**

Employee attrition is one of the most critical challenges faced by organizations as it directly impacts productivity, operational efficiency, employee morale, and overall business performance. High employee turnover increases recruitment costs, training expenses, and the time required to onboard and develop new employees. Therefore, understanding the factors that influence employee attrition is essential for organizations to improve employee retention and workforce stability.

The **HR Analytics Dashboard** was developed using the **IBM HR Employee Attrition Dataset** and Microsoft Power BI to provide a comprehensive analysis of employee behavior, workforce demographics, and attrition trends. The dashboard aims to transform raw HR data into meaningful business insights that can assist Human Resource managers and decision-makers in identifying potential causes of employee turnover and implementing effective retention strategies.

This project focuses on analyzing various employee-related factors such as age, department, job role, monthly income, overtime status, marital status, gender, job satisfaction, and years of experience within the company. By examining these variables, the dashboard helps uncover patterns and relationships that may contribute to employee attrition.

The dashboard provides a centralized platform for monitoring key workforce metrics, including total employee count, attrition count, active employees, attrition rate, average monthly income, average job satisfaction, and average years at the company. These KPIs enable management to quickly assess the overall health of the workforce and identify areas that require attention.

The project demonstrates the practical application of Business Intelligence (BI) and Data Analytics techniques in the Human Resources domain. Through data cleaning, transformation, KPI development, and visualization, the dashboard converts complex HR data into actionable insights that support evidence-based decision-making.

The primary goal of this project is not only to measure employee attrition but also to understand the underlying factors influencing employee departures. By identifying these factors, organizations can design targeted retention strategies, improve employee engagement, enhance workplace satisfaction, and reduce workforce turnover.

Overall, the HR Analytics Dashboard serves as a powerful decision-support tool that enables HR professionals and management teams to monitor workforce trends, evaluate employee retention risks, and make informed strategic decisions that contribute to long-term organizational success.

## **2. Project Objectives**

The main objectives of this dashboard are:

- Analyze overall employee attrition.
- Identify departments with high employee turnover.
- Identify job roles contributing most to attrition.
- Analyze the impact of overtime on attrition.
- Study employee demographics and workforce distribution.
- Understand relationships between salary, experience, and attrition.
- Provide actionable insights for HR decision-making.

### **3. Dataset Information**

**Dataset Name**

IBM HR Employee Attrition Dataset

**Total Records**

1,470 Employees

**Total Features**

35 Columns

### **Key Attributes**

| Attribute       | Description                    |
|-----------------|--------------------------------|
| Attrition       | Employee Left Company (Yes/No) |
| Age             | Employee Age                   |
| Department      | Department Name                |
| JobRole         | Employee Job Position          |
| MonthlyIncome   | Monthly Salary                 |
| OverTime        | Overtime Status                |
| Gender          | Male/Female                    |
| MaritalStatus   | Marital Status                 |
| JobSatisfaction | Satisfaction Score             |
| YearsAtCompany  | Years Worked in Company        |

## **4. Data Cleaning Process**

The following preprocessing steps were performed:

### **Removed Constant Columns**

- EmployeeCount
- Over18
- StandardHours

### **Data Validation**

- Checked for missing values.
- Checked duplicate records.
- Verified data types.
- Standardized column formatting.

### **Created Calculated Metrics**

- Total Employees
- Attrition Count
- Active Employees
- Attrition Rate
- Average Income
- Average Job Satisfaction
- Average Years at Company

## **5. KPIs Used**

| <b>KPI</b>               | <b>Value</b> |
|--------------------------|--------------|
| Total Employees          | 1,470        |
| Attrition Count          | 237          |
| Active Employees         | 1,233        |
| Attrition Rate           | 16.12%       |
| Average Income           | 6.5K         |
| Average Job Satisfaction | 2.73         |
| Average Years At Company | 7.01 Years   |

## **6. DAX Measures**

### **Total Employees**

Total Employees =

COUNT(Cleaned\_Data[EmployeeNumber])

### **Attrition Count**

Attrition Count = CALCULATE(

COUNT(Cleaned\_Data[EmployeeNumber]),

Cleaned\_Data[Attrition]="Yes")

### **Active Employees**

Active Employees =

[Total Employees]-[Attrition Count]

### **Attrition Rate**

Attrition Rate =

DIVIDE(

[Attrition Count],

[Total Employees])

### **Average Income**

Avg Income =

AVERAGE(Cleaned\_Data[MonthlyIncome])

### **Average Job Satisfaction**

Avg Job Satisfaction =

AVERAGE(Cleaned\_Data[JobSatisfaction])

### **Average Years At Company**

Avg Years At Company =

AVERAGE(Cleaned\_Data[YearsAtCompany])

## **7. Dashboard Visualizations**

### **KPI Cards**

Displays:

- Total Employees
- Attrition Count
- Active Employees
- Attrition Rate
- Average Income
- Average Job Satisfaction
- Average Years at Company

### **Department-wise Attrition**

#### **Visual**

Clustered Bar Chart

#### **Purpose**

Identify departments experiencing the highest employee turnover.

#### **Insight**

Research & Development and Sales departments show the highest attrition counts.

### **Job Role-wise Attrition**

#### **Visual**

Horizontal Bar Chart

#### **Purpose**

Identify job roles most affected by attrition.

#### **Insight**

Laboratory Technicians, Sales Executives, and Research Scientists contribute significantly to employee turnover.



## **Overtime Analysis**

### **Visual**

Donut Chart

### **Purpose**

Analyze employee overtime distribution.

### **Business Impact**

High overtime levels may indicate workload stress and potential burnout.

## **Marital Status Distribution**

### **Visual**

Donut Chart

### **Purpose**

Analyze workforce composition by marital status.

## **Gender Distribution**

### **Visual**

Donut Chart

### **Purpose**

Analyze workforce diversity and gender representation.

### **Insight**

The organization has a higher proportion of male employees compared to female employees.

## **Salary vs Attrition Analysis**

### **Visual**

Area Chart

### **Purpose**

Analyze salary distribution and its relationship with attrition.

### **Insight**

Lower-income employees tend to show higher attrition tendencies.

## **Experience Analysis**

### **Visual**

Line Chart

### **Purpose**

Analyze employee tenure and attrition relationship.

### **Insight**

Employees with fewer years of experience exhibit higher turnover patterns.

## **8. Key Business Insights**

### **Workforce Overview**

- Total workforce consists of 1,470 employees.
- Attrition rate is 16.12%, indicating moderate employee turnover.

### **Department Analysis**

- Research & Development experiences the highest employee attrition.
- Sales department also shows significant turnover.

### **Job Role Analysis**

- Laboratory Technicians show the highest attrition count.
- Technical and customer-facing roles are more prone to turnover.

### **Satisfaction Analysis**

- Average job satisfaction is 2.73 out of 4.
- Lower satisfaction levels may contribute to attrition.

### **Compensation Analysis**

- Employees with lower income levels tend to leave more frequently.

### **Experience Analysis**

- Employees with shorter tenure are more likely to leave the company.

## **9. Recommendations**

### **Employee Retention Strategy**

- Improve employee engagement programs.
- Conduct regular satisfaction surveys.
- Offer career growth opportunities.
- Introduce mentorship programs.

### **Work-Life Balance**

- Monitor overtime workloads.
- Promote flexible working arrangements.
- Reduce burnout risk through workload management.

### **Compensation Review**

- Evaluate salary competitiveness.
- Provide performance-based incentives.
- Introduce retention bonuses for critical roles.

### **Career Development**

- Create structured learning paths.
- Support employee skill development.
- Improve internal promotion opportunities.

## **10. Conclusion**

The HR Analytics Dashboard provides a comprehensive view of employee attrition trends within the organization. By leveraging Power BI visualizations and HR metrics, the dashboard enables management to identify attrition drivers, monitor workforce health, and implement targeted retention strategies.

The findings indicate that attrition is influenced by department, job role, compensation, overtime, and employee experience levels. Addressing these factors can significantly improve employee retention and organizational performance.